

KITSCTY AGCM, AB, Canada

WMO: 717970

Lat: 53.3533N

Lon: 110.4161W

Elev: 2178

StdP: 13.58

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-24.8	-18.8	-30.7	1.1	-24.8	-24.4	1.5	-18.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	81.6	62.9	78.5	61.7	75.6	61.0	66.8	76.7	64.8	74.5	63.0	72.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	93.8	71.8	61.0	87.0	69.6	59.2	81.3	67.8	32.6	77.3	31.0	74.5	29.6	72.5	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-30.3	87.6	5.5	3.3	-34.3	89.9	-37.5	91.9	-40.5	93.7	-44.5	96.1
			WB	-30.5	71.5	5.4	2.3	-34.4	73.2	-37.6	74.5	-40.6	75.8	-44.6	77.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.7	11.5	9.6	22.5	36.7	51.7	59.0	62.6	60.9	52.6	38.8	22.6	10.0	(6)
	DBStd	22.46	15.10	14.67	12.92	9.78	8.30	5.75	4.75	5.61	8.46	9.06	12.19	13.64	(7)
	HDD50	6171	1194	1132	852	409	81	8	0	2	68	363	823	1239	(8)
	HDD65	10408	1659	1552	1317	850	417	191	105	148	379	813	1273	1704	(9)
	CDD50	1312	0	0	0	9	132	279	391	341	145	15	0	0	(10)
	CDD65	76	0	0	0	0	4	13	31	22	6	0	0	0	(11)
	CDH74	983	0	0	0	3	107	149	309	265	146	4	0	0	(12)
	CDH80	166	0	0	0	0	11	20	52	47	36	0	0	0	(13)
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
	PrecAvg	15.9	0.7	0.5	0.6	1.1	1.7	2.9	3.1	2.2	1.3	0.7	0.6	0.7	(15)
(16) Precipitation	PrecMax	23.3	1.5	1.1	1.7	3.2	4.5	5.6	8.4	6.7	3.6	1.6	2.0	1.3	(16)
	PrecMin	8.5	0.2	0.1	0.1	0.1	0.2	0.5	0.6	0.2	0.1	0.1	0.1	0.2	(17)
	PrecStd	3.6	0.4	0.2	0.3	0.9	1.1	1.2	1.7	1.2	1.0	0.4	0.5	0.3	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.6	40.6	53.1	70.9	81.6	82.6	85.7	84.4	84.8	70.6	54.7	40.6	(19)
		MCWB	36.8	36.0	42.5	49.5	56.4	62.7	68.6	65.0	61.8	54.8	43.2	35.2	(20)
	2%	DB	38.2	36.5	46.3	63.6	77.1	78.2	80.5	80.3	78.1	61.8	46.3	36.8	(21)
		MCWB	34.4	33.2	39.0	46.6	55.1	60.3	65.6	63.5	59.2	49.6	39.7	33.1	(22)
	5%	DB	35.7	33.3	41.4	57.6	72.1	74.7	77.3	76.7	72.8	56.8	40.8	32.6	(23)
		MCWB	32.8	30.9	36.6	44.0	52.9	59.2	64.3	63.2	57.2	47.0	36.4	29.3	(24)
	10%	DB	32.1	30.6	37.8	52.1	67.9	71.3	74.2	73.5	67.4	52.1	36.9	27.9	(25)
		MCWB	29.8	28.6	34.1	41.0	51.1	57.4	62.7	61.4	54.6	44.5	33.6	25.9	(26)
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.9	36.4	44.1	50.6	58.9	66.0	71.4	69.1	63.5	55.6	45.3	36.0	(27)
		MCDB	40.9	40.2	52.3	69.6	74.6	77.1	80.7	79.8	80.1	68.9	52.7	39.8	(28)
	2%	WB	34.8	33.6	39.3	47.2	56.4	63.0	67.6	66.3	60.4	51.4	40.4	33.3	(29)
		MCDB	37.7	36.1	45.3	61.6	71.6	73.8	77.5	76.0	75.6	59.3	45.9	37.0	(30)
	5%	WB	32.7	31.2	36.6	44.5	54.5	61.1	65.6	64.3	58.0	47.9	36.7	29.5	(31)
		MCDB	35.5	33.1	40.9	56.1	69.1	70.5	75.2	73.8	70.5	55.4	40.3	32.3	(32)
	10%	WB	29.6	28.6	34.3	41.9	52.5	59.3	63.7	62.4	55.5	45.0	33.9	25.9	(33)
		MCDB	32.0	30.8	37.6	50.9	66.1	68.8	72.3	71.4	66.1	52.0	36.7	28.0	(34)
(36) Mean Daily Temperature Range	5% DB	MDBR	15.8	16.7	15.8	18.8	22.5	19.2	19.6	20.3	21.4	17.2	13.7	13.4	(35)
		MCDBR	16.1	17.9	19.1	27.9	28.1	24.6	24.6	25.9	30.3	24.9	18.9	17.2	(36)
		MCWBR	13.5	15.2	13.0	14.5	11.8	10.8	12.4	12.8	13.9	14.0	12.7	14.3	(37)
	5% WB	MCDBR	15.9	17.3	18.1	26.5	25.2	20.7	22.3	22.9	28.3	22.9	17.3	17.4	(38)
		MCWBR	13.5	15.2	12.6	14.4	11.5	10.6	12.4	12.2	13.8	13.7	12.3	14.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.242	0.291	0.281	0.326	0.347	0.366	0.362	0.372	0.309	0.274	0.248	0.217	(40)
	taud		2.333	2.195	2.394	2.377	2.365	2.339	2.377	2.326	2.484	2.552	2.452	2.361	(41)
	Ebn at Noon		243	255	291	288	286	280	279	269	277	268	242	235	(42)
	Edh at Noon		17	28	29	34	36	37	36	35	26	20	16	14	(43)
(44) All-Sky Solar Radiation	RadAvg		315	651	1127	1490	1847	1886	1954	1593	1122	651	314	234	(44)
	RadStd		33	47	89	132	130	136	107	120	122	92	34	23	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page